Akoizo Original Practice - 2027 Rotation Form 1

Division B | Meteorology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which satellite channel can show a jet stream in clear air?

Unit: Observing severe weather: stations, balloons, radar, and satellites.
Worked-case data: A radar image shows a storm with reflectivity of 65 dBZ, a tight red-green velocity couplet on its southwest side, and a small area of CC values around 0.7 at the same spot. What is happening?
Separate investigation observations:
Reflectivity: A hook-shaped appendix on the storm's southwest side.
Velocity: Strong inbound and outbound winds side by side at the hook.
Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. Visible

B. Infrared only at night

C. Radar reflectivity

D. Water vapor

Answer: _____

2. (2 points) Unsaturated air rises and cools at about:

Unit: Stability on Stüve and Skew-T diagrams.
Worked-case data: The surface temperature is 30 °C and the dew point is 22 °C. Estimate the LCL and comment on tornado potential.
Separate investigation observations:
Surface: Temperature 32 °C, dew point 23 °C.
Layer near 850 mb: A thin warm layer: temperature rises with height.
Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. 9.8 °C per km

B. 2 °C per km

C. 6.5 °C per km always

D. 20 °C per km

Answer: _____

3. (2 points) Which feature defines a supercell?

Unit: Thunderstorms and lightning.
Worked-case data: You see a flash and hear thunder 12 seconds later. How far away is the lightning, and are you safe outdoors?
Separate investigation observations:
Duration: The storm has lasted three hours.
Radar: A hook echo and a bounded weak-echo region.
Velocity: Persistent rotation in the mid levels.
Reports: Hail up to 2.75 inches (baseball size).

A. A warm front

B. A rotating updraft (mesocyclone)

C. A gust front

D. A single short-lived cell

Answer: _____

4. (2 points) A hurricane with sustained winds of 115 mph is:

Unit: Hurricanes, typhoons, and cyclones.
Worked-case data: A hurricane with 140 mph sustained winds approaches a coast with a wide, shallow continental shelf at high tide. Rate the storm and assess surge.
Separate investigation observations:
Satellite: A clear eye surrounded by a ring of very cold cloud tops.
Recon aircraft: Sustained surface winds of 160 mph.
Motion: Slowing to 5 mph near the coast.
Coast: A wide, shallow bay ahead of the storm.

A. Category 4

B. Category 1

C. Category 3

D. Category 2

Answer: _____

5. (2 points) On Doppler velocity, green colors usually mean:

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Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. Heavy rain

B. Hail

C. Motion toward the radar

D. Motion away from the radar

Answer: _____

6. (2 points) On a sounding, temperature and dew point lines close together indicate:

Unit: Stability on Stüve and Skew-T diagrams.
Worked-case data: The surface temperature is 30 °C and the dew point is 22 °C. Estimate the LCL and comment on tornado potential.
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Surface: Temperature 32 °C, dew point 23 °C.
Layer near 850 mb: A thin warm layer: temperature rises with height.
Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. Very dry air

B. An inversion

C. Strong wind

D. Nearly saturated air

Answer: _____

7. (2 points) In the dissipating stage of a single-cell storm:

Unit: Thunderstorms and lightning.
Worked-case data: You see a flash and hear thunder 12 seconds later. How far away is the lightning, and are you safe outdoors?
Separate investigation observations:
Duration: The storm has lasted three hours.
Radar: A hook echo and a bounded weak-echo region.
Velocity: Persistent rotation in the mid levels.
Reports: Hail up to 2.75 inches (baseball size).

A. The downdraft dominates

B. The updraft strengthens

C. The cloud is still growing

D. Rain has not begun

Answer: _____

8. (2 points) Where are a hurricane’s strongest winds?

Unit: Hurricanes, typhoons, and cyclones.
Worked-case data: A hurricane with 140 mph sustained winds approaches a coast with a wide, shallow continental shelf at high tide. Rate the storm and assess surge.
Separate investigation observations:
Satellite: A clear eye surrounded by a ring of very cold cloud tops.
Recon aircraft: Sustained surface winds of 160 mph.
Motion: Slowing to 5 mph near the coast.
Coast: A wide, shallow bay ahead of the storm.

A. Far ahead of the storm

B. In the eyewall

C. In the eye

D. In the outer rain bands

Answer: _____

9. (2 points) Why are cold, bright cloud tops on infrared images important?

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Correlation coefficient: A small area of 0.6-0.7 inside the hook.

A. They indicate clear skies

B. They mark tall thunderstorms

C. They mark fog

D. They show surface temperature

Answer: _____

10. (2 points) An environmental lapse rate between the moist and dry rates is:

Unit: Stability on Stüve and Skew-T diagrams.
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Separate investigation observations:
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Above 700 mb: Temperature drops steeply with height.
Afternoon: A front lifts the air by midafternoon.

A. Absolutely unstable

B. Always neutral

C. Conditionally unstable

D. Absolutely stable

Answer: _____

11. (2 points) Interpret the adjacent opposite radial velocities cautiously.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

12. (2 points) What additional interpretation is supported by low correlation coefficient collocated with the velocity couplet?

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

13. (2 points) Calculate the surface temperature-dew-point spread.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

14. (2 points) Identify one reason that 65 dBZ reflectivity alone does not prove a tornado.

Synthetic diagnostic table (not a real warning):
Reflectivity | 65 dBZ
Velocity | Adjacent inbound/outbound maxima
Correlation coefficient | 0.70, collocated with couplet
Surface temperature | 26 C
Dew point | 21 C
Wind | Sharp shift across boundary

Answer: _____
_____
_____

15. (2 points) A storm moves 25 km east in 30 minutes. Find mean speed.

Storm-track table:
Time | Eastward distance
0 min | 0 km
30 min | 25 km

Answer: _____
_____
_____

16. (2 points) A rain gauge records 35 mm in 0.5 hours. Calculate mean rain rate.

Gauge table:
Elapsed time | Accumulated rain
0 h | 0 mm
0.5 h | 35 mm

Answer: _____
_____
_____

17. (2 points) A satellite IR time series shows cloud-top brightness temperature falling from 240 K to 220 K. Give a physically plausible interpretation and a limitation.

IR table:
Frame 1 | 240 K
Frame 2 | 220 K

Answer: _____
_____
_____

18. (2 points) Station wind shifts form a coherent boundary while pressure rises behind it. Describe a candidate thunderstorm-related mechanism.

Station table:
Ahead of boundary | Original wind, lower pressure
Behind boundary | Shifted wind, rising pressure

Answer: _____
_____
_____

19. (2 points) Radar detects strong winds perpendicular to its beam but a small radial velocity. Explain why the wind need not be weak.

Radar geometry table:
Wind direction | Perpendicular to beam
Radial velocity | Small
Independent wind observation | Strong

Answer: _____
_____
_____

20. (2 points) A short intense rain period falls on an already saturated basin. Explain why rainfall amount alone does not establish flood severity.

Basin table:
Antecedent soil | Saturated
Rain | Short, intense burst
Catchment drainage/routing | Not supplied

Answer: _____
_____
_____